Sampling & Data Collection

Exam code: 9MA0

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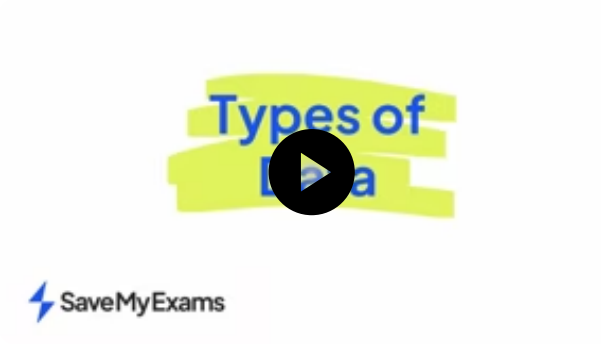

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Types of data

What are the different types of data?

- **Qualitative** data is data that is usually given in words not numbers to **describe** something
  - For example: the colour of a teacher's car
- **Quantitative** data is data that is given using numbers which **counts or measures** something
  - For example: the number of pets that a student has
- **Discrete** data is quantitative data that needs to be **counted**
  - Discrete data can only take **specific values** from a set of (usually finite) values
  - For example: the number of times a coin is flipped until a tails is obtained
- **Continuous** data is quantitative data that needs to be **measured**
  - Continuous data can take **any value** within a range of infinite values
  - For example: the height of a student
- **Age** can be **discrete or continuous** depending on the context or how it is defined
  - If you mean how many years old a person is then this is discrete

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

What other key words do I need to know?

Help me understand

- The **population** refers to the **whole set** of things which you are interested in
  - For example: if a vet wanted to know how long a typical French bulldog slept for in a day then the population would be all the French bulldogs in the world
- A **sample** refers to a **subset of the population** which is used to collect data from
  - For example: the vet might take a sample of French bulldogs from different cities and record how long they sleep in a day
- A **sampling frame** is a **list** of all members of the **population**
  - For example: a list of employees' names within a company
- A **population parameter** is a **numerical value** which describes a **characteristic of the population**
  - These are usually unknown
  - For example: the mean height of all 16-year-olds in the UK
- A **sample statistic** is a **value** computed using **data from the sample**
  - These are used to estimate population parameters
  - For example: the mean height of 200 16-year-olds from randomly selected cities in the UK

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Sampling techniques

What are the differences between a census and sampling?

- A **census** collects data about **all** the members of a **population**

- The main advantage of a census is that it gives fully accurate results
- The disadvantages of a census are:
  - It is time consuming and expensive to carry out
  - It can destroy or use up all the members of a population when they are consumables (imagine a company testing every single firework)

- **Sampling** is used to collect data from a **subset of the population**
- The advantages of sampling are:
  - It is quicker and cheaper than a census
  - It leads to less data needing to be analysed

- The disadvantages of sampling are:
  - It might not represent the population accurately
  - It could introduce bias

What sampling techniques do I need to know?

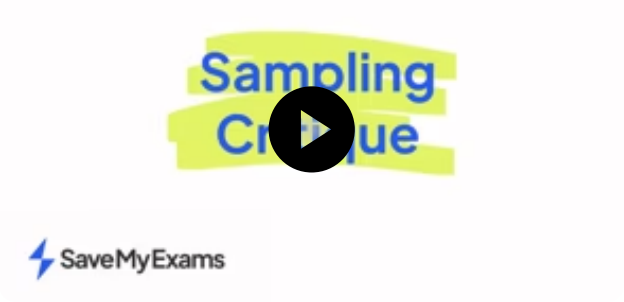
- **Simple random sampling**: if a sample of size *n* is taken then every group of *n* members from the population has an equal probability of being selected for the sample
  - Simple random sampling is carried out by uniquely numbering every member of a population and randomly selecting n different numbers using a random number generator or a form of lottery (where numbers are selected randomly)

- **Systematic sampling**: a sample is formed by choosing members of a population at regular intervals using a list
  - To carry this out you would calculate the size of the interval $k = \frac{size\ of\ population\ (N)}{size\ of\ sample\ (n)}$ and choose a starting point between 1 and *k* then select every *k*th member after the first one

- **Stratified sampling**: the population is divided into disjoint groups (called strata) and then a random sample is taken from each group (stratum)
  - The proportion of a sample that belongs to a stratum is equal to the proportion of the population that belongs to the stratum
  - The number of members sampled from a stratum $= \frac{size\ of\ sample\ (n)}{size\ of\ population\ (N)}$ x number of members in the stratum
  - The population could be split by age ranges, gender, etc

- **Quota sampling**: the population is split into groups (like stratified sampling) and members of the population are selected until each quota is filled
  - If a member does not want to be included then another member is chosen instead
  - The members do not have to be selected randomly

- **Opportunity (convenience) sampling**: a sample is formed using available members of the population who fit the criteria



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Sampling critique

When should each sampling technique be used or avoided?

- **Simple random sampling**: this should be used when you want a **random sample** to avoid bias
  - Useful when you have a small population or want a small sample (such as children in a class)
  - This can not be used if it is not possible to number or list all the members of the population (such as fish in a lake)
- **Systematic sampling**: this should be used when you want a **random sample** from a **large population**
  - Useful when there is a natural order (such as a list of names or a conveyor belt of items)
  - In order for the sample to be random the sampling frame needs to be random
  - This can not be used if it is not possible to number or list all the members of the population (such as penguins in Antarctica)
- **Stratified sampling**: this should be used when the population can be **split into obvious groups** of members (where members within a group have a common characteristic)
  - Useful when there are very different groups of members within a population
  - The sample will be representative of the population structure
  - The members selected from each stratum are chosen randomly
  - This can not be used if the population can not be split into groups or if the groups overlap
- **Quota sampling**: this should be used when a small sample is needed to be **representative** of the population structure
  - Useful when collecting data by asking people who walk past you in a public place or when a sampling frame is not available

- **Opportunity (convenience) sampling**: this should be used when a sample is needed quickly
  - Useful when a list of the population is not possible
  - This is unlikely to be representative of the population structure

What are the main criticisms of sampling techniques?

- Most sampling techniques can be improved by taking a larger sample
- Sampling can introduce bias - so you want to minimise the bias within a sample
  - To minimise bias the sample should be random
- A sample only gives information about those members
  - Different samples may lead to different conclusions about the population

Worked Example

Mike is a biologist studying mice in an open enclosure. He has access to approximately 540 field mice and 260 harvest mice. Mike wants to sample 10 mice and he wants the proportions of the two types of mice in his sample to reflect their respective proportions of the population.

(a) Calculate the number of field mice and harvest mice that Mike should include in his sample.

(b) Given that Mike does not have a list of all mice in the enclosure, state the name of this sampling method.

(c) Suggest one way in which Mike could improve his sampling method.

Answer:

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about the sampling methods that have been used.

- Stratified and quota sampling seem similar, but the main difference is stratified involves randomly selecting the members within each stratum.

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